



University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab

LAB REPORT

Experiment Number: 02

Experiment Title: Traffic Light Control using Servo Motor and LEDs

Submitted by:

Name : Sunandan Das

Student ID : 23101218

Section : D2

Submitted to:

Zaima Sartaj Taheri

Lecturer,

Department of Computer Science and Engineering

Date of Submission: 27/02/2026

1. Experiment Name

Traffic Light Control using Servo Motor and LEDs

2. Objective

The main objective of this experiment is to design and implement a microcontroller-based traffic light control system using an Arduino Uno, with an additional servo motor acting as a barrier or gate. The system simulates how real-world traffic signals operate at road crossings or parking entrances.

Specific Objectives:

- To learn how to interface LEDs and a servo motor with the Arduino Uno.
- To implement sequential control of Red, Yellow, and Green lights using delay-based timing.
- To synchronize the servo motor with the traffic light signals to simulate a barrier control system.
- To gain hands-on practice in coding, circuit design, and real-time simulation of traffic management.

3. Apparatus / Hardware & Software Requirements

• **Microcontroller:** 1 × Arduino Uno R3

• **Actuators:**

- 3 × LEDs (Red, Yellow, Green)

- 1 × Servo Motor

• **Software:**

- Arduino IDE

- Tinker CAD

• **Other Components:**

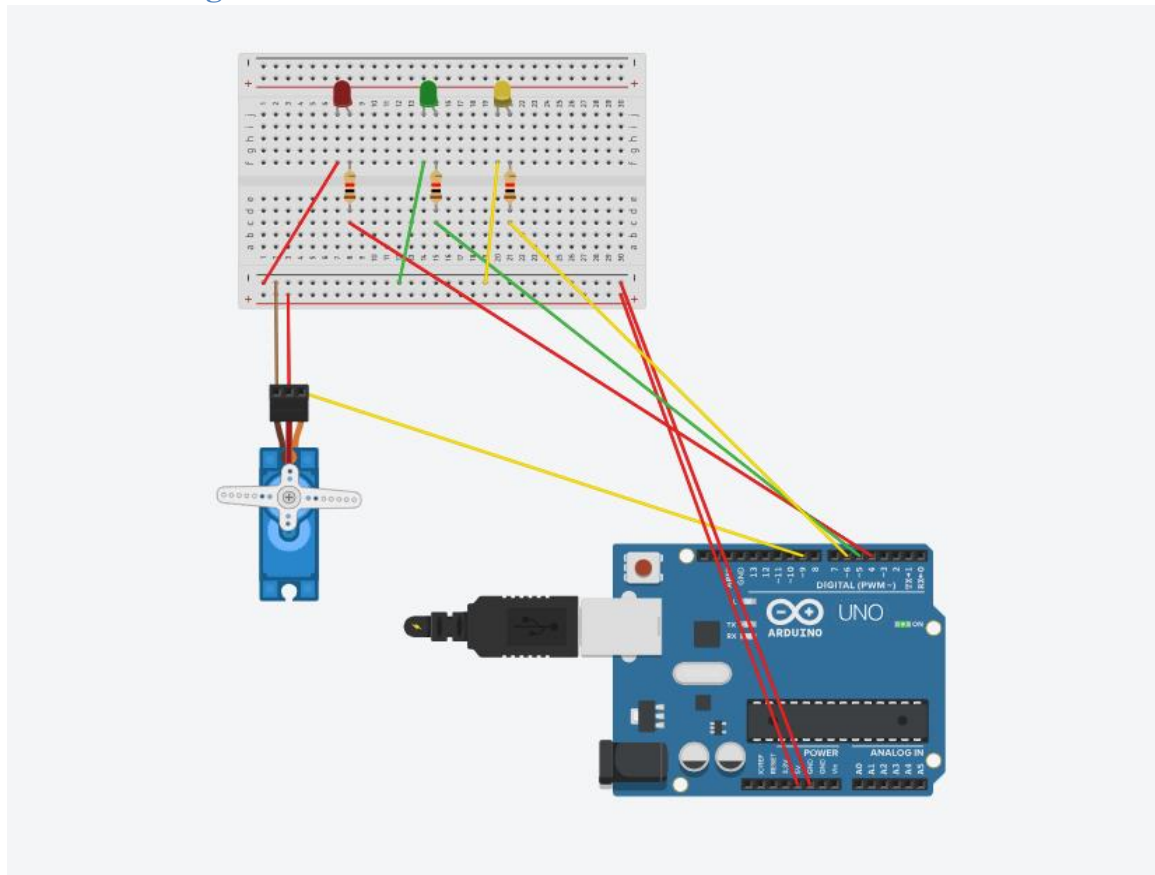
- 1 × Breadboard

- Jumper Wires

- 3 × Resistors

- 1 × USB Cable

4. Circuit Diagram / Schematic



5. Code / Assembly Program

```
#include <Servo.h>

Servo barrierServo;

int redLED = 4;

int yellowLED = 5;

int greenLED = 6;

void setup() {

  barrierServo.attach(9);

  pinMode(redLED, OUTPUT);
```

```
pinMode(yellowLED, OUTPUT);  
pinMode(greenLED, OUTPUT);  
}
```

```
void loop() {  
    // Green ON  
  
    digitalWrite(greenLED, HIGH);  
    digitalWrite(yellowLED, LOW);  
    digitalWrite(redLED, LOW);  
    barrierServo.write(90);  
    delay(5000);  
  
    // Yellow ON  
  
    digitalWrite(greenLED, LOW);  
    digitalWrite(yellowLED, HIGH);  
    digitalWrite(redLED, LOW);  
    delay(2000);  
  
    // Red ON  
  
    digitalWrite(greenLED, LOW);  
    digitalWrite(yellowLED, LOW);  
    digitalWrite(redLED, HIGH);  
    barrierServo.write(0);  
    delay(5000);  
}
```

6. Output / Observations

After uploading the program to the Arduino Uno and connecting the LEDs and servo motor, the system worked as a traffic light controller with an automated barrier gate.

The behavior observed is as follows:

1. Green Signal

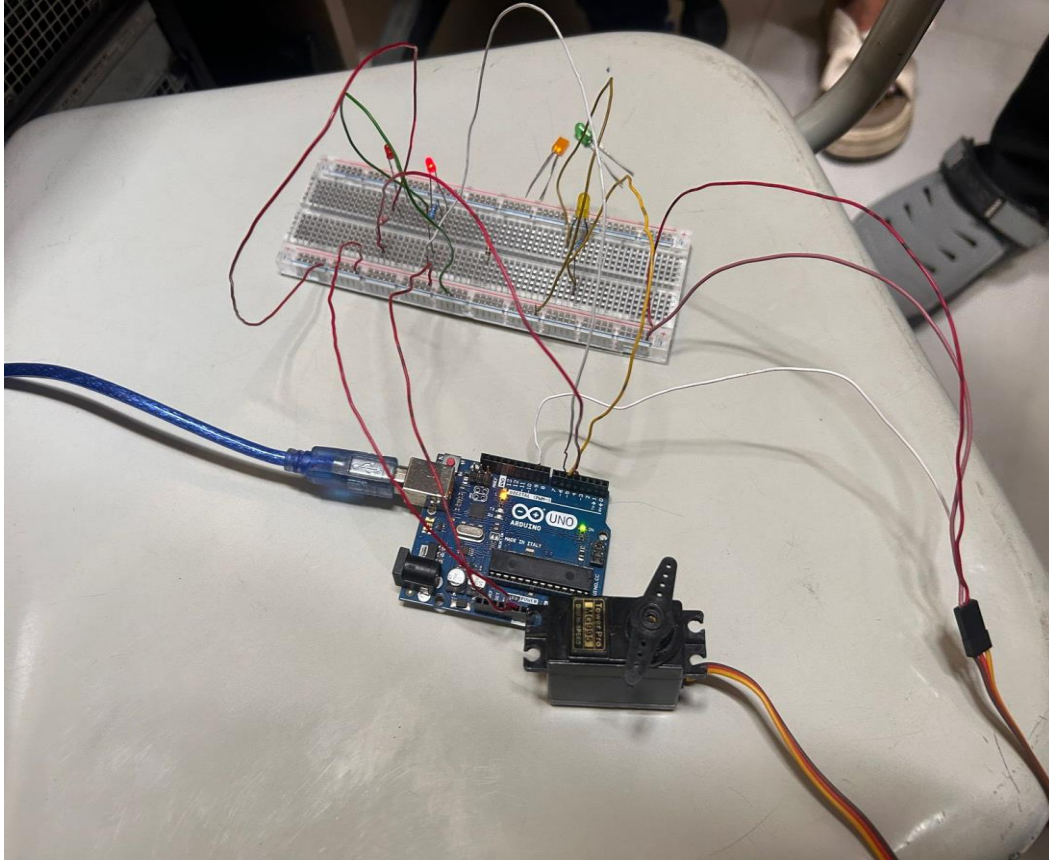
- Green LED turns ON (Red and Yellow LEDs remain OFF)
- Servo motor rotates to 90° → Barrier opens
- This state remains for 5 seconds

2. Yellow Signal

- Yellow LED turns ON (Red and Green LEDs remain OFF)
- Servo motor remains at 90° → Barrier stays open
- This state remains for 2 seconds

3. Red Signal

- Red LED turns ON (Green and Yellow LEDs remain OFF)
- Servo motor rotates to 0° → Barrier closes
- This state remains for 5 seconds



7. Result

The Arduino-based traffic light system was successfully implemented. The Red, Yellow, and Green LEDs operated sequentially, and the servo motor functioned as a barrier gate, opening at Green and Yellow (90°) and closing during Red (0°). The entire sequence repeated continuously, simulating a real traffic signal with an automated barrier control.

During the experiment, an issue was encountered where one of the LED lights did not function properly. The program code and all circuit connections were carefully checked and found to be correct. The faulty LED was then replaced; however, the problem persisted. After replacing the LED a second time, the third LED worked correctly. Once the issue was resolved, the entire system operated successfully and performed as expected.

8. Conclusion

This experiment effectively demonstrated a traffic light control system using an Arduino Uno, with a servo motor incorporated to function as a barrier gate. The Red, Yellow, and Green LEDs operated sequentially according to the programmed timing, while the servo motor opened at 90° during the Green and Yellow signals and closed at 0° during the Red signal, effectively simulating a real-world traffic signal with an automated barrier. During the experiment, a faulty LED was identified and replaced, and once the issue was resolved, the system functioned smoothly and reliably. Overall, this experiment provided valuable hands-on experience in Arduino programming, circuit assembly, component interfacing, troubleshooting, and real-time traffic control simulation, enhancing practical understanding of how microcontrollers can be applied in automated traffic management systems.